

Dimensionless Numbers

The ratios that govern microfluidic flow — a Semitree cheat sheet

At the microscale, a handful of dimensionless numbers tell you which forces win and how a device behaves. Each is a ratio of two competing effects.

Reynolds number

Re

$$Re = \rho \cdot v \cdot Dh / \mu$$

inertial vs. viscous forces — Laminar when $Re \ll 2000$; microchannels sit far below this.

Péclet number

Pe

$$Pe = v \cdot L / D$$

advection vs. diffusion — High Pe: flow carries species faster than diffusion can mix them.

Capillary number

Ca

$$Ca = \mu \cdot v / \gamma$$

viscous vs. interfacial forces — Sets the droplet-formation regime in two-phase flow.

Weber number

We

$$We = \rho \cdot v^2 \cdot L / \gamma$$

inertial vs. interfacial forces — Predicts droplet breakup and jetting.

Bond number

Bo

$$Bo = \Delta\rho \cdot g \cdot L^2 / \gamma$$

gravity vs. interfacial forces — Tiny at the microscale — surface tension dominates gravity.

Capillary length

Lc

$$Lc = \sqrt{\gamma / (\Delta\rho \cdot g)}$$

where gravity $\approx$ surface tension — Below this length scale, capillarity rules (~2.7 mm for water/air).

Variables

ρ density (kg/m³)

v mean velocity (m/s)

Dh hydraulic diameter (m)

μ dynamic viscosity (Pa·s)

L characteristic length (m)

D diffusion coefficient (m²/s)

γ surface tension (N/m)

$\Delta\rho$ density difference (kg/m³)

g gravity (9.81 m/s²)